

Electrical Contact Pressure Gauge (EP)

Electrical Contact Pressure Gauge (EP) with electrical contacts are suitable for controlling and regulating process pressure parameters. The contact opens and closes electrical circuits in relation to the position of the pointer on the pressure gauge.

The use of corrosion resistant material such as high grade stainless steel for both the casing and the measurement system makes this pressure gauge best suited to operate in an adverse environment and with corrosive media. The tried and tested bourdon tube system coupled with modular contacts provides reliable control of operating pressure.

Electrical contacts are used as snap action switches especially designed for harsh industrial conditions and the choice of different electrical contact material enables high currents to be switched reliably. If the electrical switching capacities of the contact are exceeded or not reached, a relay is to be used to provide appropriate current rating.

The welded design coupled with modular construction system allows for best-in-class ingress protection (upto IP-65 standard), also, offers a long and stable service life of the gauge with tried and tested safety features such as vent plug adhering to EN 837-1 standard.



Available In

Nominal sizes	4"(100mm) and 6"(150mm)
Ranges	-1 to 0 Kg/Sq.cm and 0 to 700 Kg/Sq.cm

Special Features

Compliance to latest EN 837-1

Standard ranges : -1 to 0 Kg/Sq.cm & 0 to 700 Kg/Sq.cm

Overpressure limit 125 % of FSD without contact, 100% with contact

Accuracy class $\pm 1.0\%$ of FS (contact accuracy 2.5%)

Ingress protection to IP65

Dry or liquid filled version

CE marking as per PED 97/23/EC

Stainless steel casing

Socket to case – TIG argon welded

Industrial Applications

Chemical and fertiliser

Pharmaceutical

Food processing

Water treatment plants

Petrochemical

Thermal, nuclear and hydroelectric power stations

Iron and steel

Rubber moulding / processing plants

Hydraulics and pneumatics

Hydraulic power packs

Cryogenics

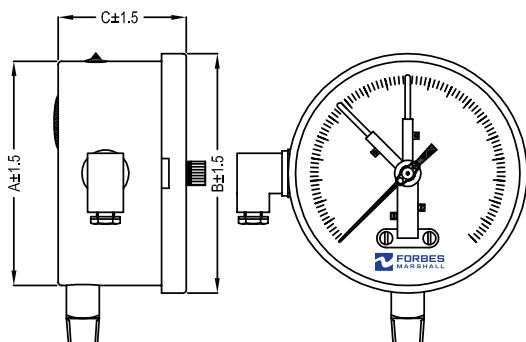
Refineries

Standard Product Specifications	
Specifications	Description
Reference standard	EN 837 - 1
Mounting	Direct bottom
Connection	½” NPT (M)
Case and bezel	SS 304
Bourdon	SS 316L
Socket	SS 316L
Movement	SS 304
Contact with rating	1 NO 1 NC , 1 Amp 230 VAC
Dial	Aluminum, white background with black marking
Pointer	Aluminum, black color micrometer with zero adjustable
Accuracy	±1% of FS (standard) (contact accuracy 2.5%)
Overpressure limit	As per EN standard
	From 0 to 100 Kg/Sq.cm 125% of maximum scale value (without contact)
	From 100 to 600 Kg/Sq.cm 115% of maximum scale value (without contact)
Window	Polycarbonate glass
Protection	IP65
Blow off disc and filling plug	Neoprene
Gasket	Silicon
Ambient temperature	-25°C to 65°C
Process temperature	Max. 120°C (for dry gauge)
	Max. 100°C (for di-electric silicon oil filled gauges)

Mounting Options

Filled Version

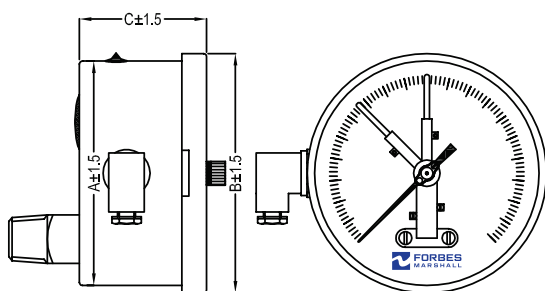
Direct Bottom



→ Parameters

Dial Size	A	B	C
100	Ø100	Ø110	90.0
150	Ø150	Ø160	90.0

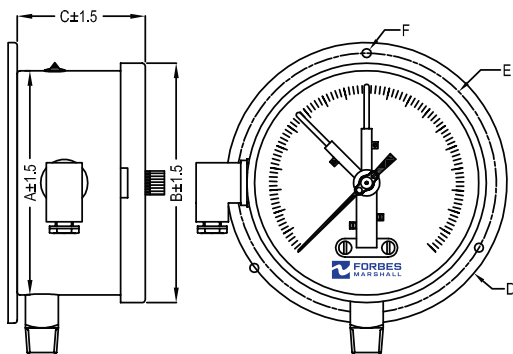
Lower Back



→ Parameters

Dial Size	A	B	C
100	Ø100	Ø110	90.0
150	Ø150	Ø160	90.0

Surface Bottom

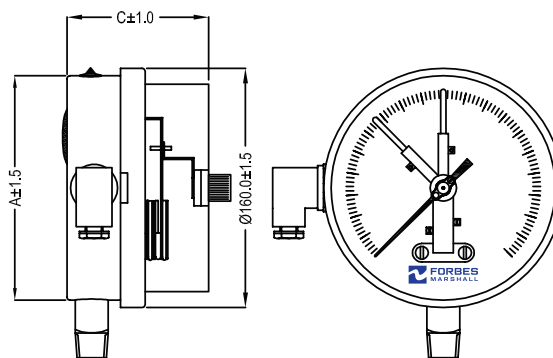


→ Parameters

Dial Size	A	B	C	D	E	F
100	Ø100	Ø110	90.0	Ø134	Ø118	Ø6
150	Ø150	Ø160	90.0	Ø186	Ø168	Ø6

Dry Version

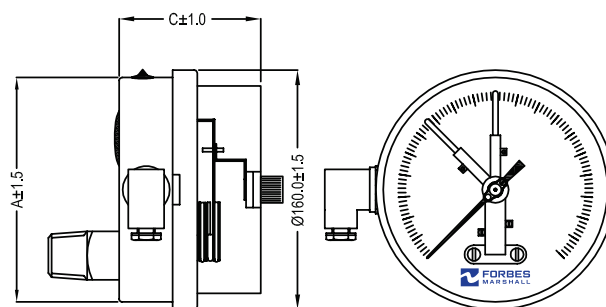
Direct Bottom



→ Parameters

Dial Size	A	B	C
100	Ø100	Ø110	90.0
150	Ø150	Ø160	90.0

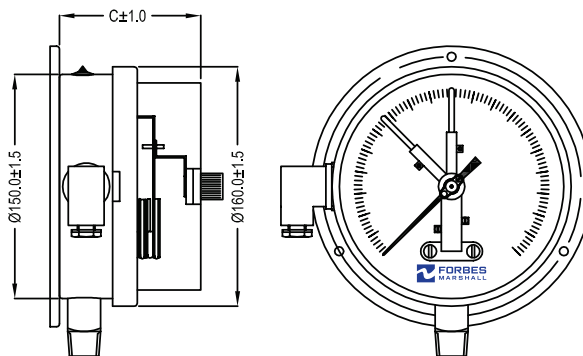
Lower Back



→ Parameters

Dial Size	A	B	C
100	Ø100	Ø110	90.0
150	Ø150	Ø160	90.0

Surface Bottom



→ Parameters

Dial Size	A	B	C	D	E	F
100	Ø100	Ø110	90.0	Ø134	Ø118	Ø6
150	Ø150	Ø160	90.0	Ø186	Ø168	Ø6

Pressure Ranges

Standard Pressure Ranges					
Code	Kg/Sq.cm (Code A)	Bar (Code B)	Code	KPA (Code C)	
AA	-1 + 0	-1 + 0	AR	-100 + 0	
AB	0 + 0.6	0 + 0.6	AL	0 + 60	
AC	0 + 1	0 + 1	AM	0 + 100	
AD	0 + 1.6	0 + 1.6	AN	0 + 160	
AE	0 + 2.5	0 + 2.5	AO	0 + 250	
AF	0 + 4	0 + 4	AP	0 + 400	
AG	0 + 6	0 + 6	AQ	0 + 600	
AH	0 + 10	0 + 10	AY	0 + 1000	
AI	0 + 16	0 + 16	AZ	0 + 1600	
AJ	0 + 25	0 + 25	BA	0 + 2500	
AK	0 + 40	0 + 40	BB	0 + 4000	
AL	0 + 60	0 + 60	BC	0 + 6000	
AM	0 + 100	0 + 100	BD	0 + 10000	
AN	0 + 160	0 + 160	BE	0 + 16000	
AO	0 + 250	0 + 250	BF	0 + 25000	
AP	0 + 400	0 + 400	BG	0 + 40000	
AQ	0 + 600	0 + 600	BH	0 + 60000	
Code	Dual Scale kg/Sq.cm & Psi (Code D)	Dual Scale Bar & Psi (Code E)	Code	Dual Scale kg/Sq.cm & Psi (Code D)	Dual Scale Bar & Psi (Code E)
BI	-1 + 0/-15 + 0	-1 + 0/-15 + 0	CI	0 + 210/0 + 3000	0 + 210/0 + 3000
BK	0 + 1/0 + 15	0 + 1/0 + 15	CJ	0 + 280/0 + 4000	0 + 280/0 + 4000
BL	0 + 1.6/0 + 23	0 + 1.6/0 + 23	CK	0 + 420/0 + 6000	0 + 420/0 + 6000
BZ	0 + 2.1/0 + 30	0 + 2.1/0 + 30	CL	0 + 700/0 + 10000	0 + 700/0 + 10000
CA	0 + 4.2/0 + 60	0 + 4.2/0 + 60			
BO	0 + 6.0/0 + 86	0 + 6.0/0 + 86			
CB	0 + 7.0/0 + 100	0 + 7.0/0 + 100			
BP	0 + 10/0 + 150	0 + 10/0 + 143			
CC	0 + 14.0/0 + 200	0 + 14.0/0 + 200			
BQ	0 + 16.0/0 + 230	0 + 16.0/0 + 230			
CD	0 + 21.0/0 + 300	0 + 21.0/0 + 300			
CE	0 + 28.0/0 + 400	0 + 28.0/0 + 400			
CF	0 + 42.0/0 + 600	0 + 42.0/0 + 600			
CG	0 + 70.0/0 + 1000	0 + 70.0/0 + 1000			
BU	0 + 100/0 + 1500	0 + 100/0 + 1500			
CH	0 + 140/0 + 2000	0 + 140/0 + 2000			
BV	0 + 160/0 + 2300	0 + 160/0 + 2300			

Compound Pressure Range

Code	mmHg / Kg/Sq.cm (Code F)	mmHg/Bar (Code G)	Code	Kg/Sq.cm Code A)	Bar (Code B)
CM	-760 + 1.5	-760 + 1.5	CU	-1 + 1.5	-1 + 1.5
CN	-760 + 3.0	-760 + 3.0	CV	-1 + 3.0	-1 + 3.0
CO	-760 + 5.0	-760 + 5.0	CW	-1 + 5.0	-1 + 5.0
CP	-760 + 6.0	-760 + 6.0	CX	-1 + 6.0	-1 + 6.0
CQ	-760 + 7.0	-760 + 7.0	CY	-1 + 7.0	-1 + 7.0
CR	-760 + 9.0	-760 + 9.0	CZ	-1 + 9.0	-1 + 9.0
CS	-760 + 15.0	-760 + 15.0	DA	-1 + 15.0	-1 + 15.0
CT	-760 + 24.0	-760 + 24.0	DB	-1 + 24.0	-1 + 24.0

Code	Dual Scale in Hg+ Psi / mm Hg+Kg/Sq.cm (Code H)	Dual Scale in Hg + Psi / mm (Code I)	Code	Dual Scale in Hg + Psi Kg/Sq.cm (Code J)	Dual Scale in Hg +Psi / Bar (Code K)
DC	-30 + 20/-760 + 1.5	-30 + 20/-760 + 1.5	FA	-30 + 20/-1 + 1.5	-30 + 20/-1 + 1.5
DD	-30 + 42/-760 + 3.0	-30 + 42/-760 + 3.0	FB	-30 + 42/-1 + 3.0	-30 + 42/-1 + 3.0
DE	-30 + 70/-760 + 5.0	-30 + 70/-760 + 5.0	FC	-30 + 70/-1+ 5.0	-30 + 70/-1+ 5.0
DF	-30 + 85/-760 + 6.0	-30 + 85/-760 + 6.0	FD	-30 + 85/-1 + 6.0	-30 + 85/-1 + 6.0
DG	-30 + 100/-760 + 7.0	-30 + 100/-760 + 7.0	FE	-30 + 100/-1 + 7.0	-30 + 100/-1 + 7.0
DH	-30 + 125/-760 + 9.0	-30 + 125/-760 + 9.0	FF	-30 + 125/-1 + 9.0	-30 + 125/-1 + 9.0
DI	-30 + 210/-760 + 15.0	-30 + 210/-760 + 15.0	FG	-30 + 210/-1 + 15.0	-30 + 210/-1 + 15.0
DJ	-30 + 340/-760 + 24.0	-30 + 340/-760 + 24.0	FH	-30 + 340/-1 + 24.0	-30 + 340/-1 + 24.0

Ordering Information

		**	*	*	*	*	**	*	*		
Model		EP									
Dial Size											
100 mm (4")	E										
150 mm (6")	F										
Mountings											
Direct bottom	1										
Lower back	2										
Surface bottom	4										
Connection Size											
1/4"	A										
3/8"	B										
1/2"	C										
M20X1.5P	1										
Connection type											
NPT (M)	A										
BSP (M)	B										
BSPT (M)	C										
Metric thread (M)	M										
Range											
Refer to Std. Pressure Range Table											
Engineering Unit**											
Kg/sq.cm	A										
Bar	B										
KPA	C										
Dual scale Kg/sq.cm-Psi	D										
Dual scale Bar-Psi	E										
CONTACT											
1 NO Single contact	A										
1 NC Single contact	B										
1 NO 1 NC Double contact	C										
1 NC 1 NO Double contact	D										
2 NO Double contact	E										
2 NC Double contact	F										
1 SPDT Single pole double throw	G										
Others please specify											
Case and Bezel											
SS304 (standard)											
SS316											
Bourdon and Socket											
SS316L (standard)											
Monel											
Movement											
SS304 (standard)											
SS316											
Pointer											
Micrometer pointer (standard)											
Window											
Polycarbonate glass (standard)											
Acrylic (for filled version)											
Optional Features											
Accuracy 1 % (contact accuracy 2.5%) (standard)											
Di-electric silicon oil case filling											
5 Amps. contact rating											
SS tag plate											
Custom designed dial											
Mirror scale dial											
Seamless bourdon											
Tests and Certificates											
5 point calibration (standard)											
Material test certificate											
NACE conformity to wetted parts											
Oxygen service											
EN 10204 3.1 B certification											
Third party inspection											
Specialised requirement (please specify)											

Ordering example: EPE1CAACAA-0000 (with standard selection)



Forbes Marshall
Krohne Marshall
Forbes Marshall Arca
Codel International
Forbes Solar
Forbes Vyncke
Forbes Marshall Steam Systems

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